

Contact

akshithashetty982@gmail.com

www.linkedin.com/in/akshitha-shetty-01a1a2284 (LinkedIn)

Top Skills

ROS 2 (Humble)

Autonomous Guided Vehicles (AGV)

SOLIDWORKS

Certifications

ISRO Bharatiya Antariksh Hackathon 2026 – Participant

Simulink Onramp

Matlab Onramp

Akshitha Shetty

Final Year Robotics & Automation Engineer | NITK Intern | Skilled in ROS2, MATLAB & SolidWorks | Seeking Full-Time Roles
Mangaluru, Karnataka, India

Summary

I am a final-year Robotics and Automation Engineering student at Sahyadri College of Engineering & Management, deeply passionate about building intelligent systems that bridge the gap between software algorithms and physical hardware. From developing autonomous warehouse robots to designing UAV launch systems, I thrive on solving complex, multi-disciplinary engineering challenges. Throughout my academic journey and as an active core member of Team Challengers, I have cultivated a hands-on approach to robotics. I believe true innovation comes from continuous learning, consistency, and a drive to build things that work in the real world, not just in simulation. My technical expertise spans the entire robotics stack: Autonomous Systems & Software: Proficient in ROS 2 (Humble), Nav2, SLAM Toolbox, and Gazebo. I have engineered hybrid navigation pipelines for AMRs and dynamic task-scheduling algorithms for warehouse picking robots (MAHE Hackathon 2026). UAVs & Aerospace: Interned at NITK, where I designed catapult launch and parachute recovery systems for autonomous UAVs. I also contributed to lunar rover path-planning algorithms for the ISRO BAH Hackathon and advanced to the Elimination Round of the ISRO Robotics Challenge (ASCEND). Computer Vision & AI: Applied OpenCV, YOLO, and machine learning to solve real-world problems, ranging from agricultural water management (Smart India Hackathon 2024) to real-time ArUco marker detection for spatial awareness. Hardware & Mechanical Design: Skilled in SolidWorks, kinematics, and rapid prototyping. I have designed custom multi-axis robotic arms and successfully integrated low-level hardware using Raspberry Pi, Arduino, and Pixhawk flight controllers. I am actively seeking full-time roles in robotics, autonomous systems, and integration engineering where I can contribute to pushing the boundaries of what machines can do. Let's connect and build the future of autonomy!

Experience

Team Challengers

Team member

June 2024 - Present (2 years 4 months)

Mangaluru

Active contributor to a prominent student-led aviation and aerospace club, involved in end-to-end technical fabrication and engineering projects since freshman year.

Designed, built, and piloted various aeromodels, representing the institution and competing at premier national technical events, including IIT Kanpur.

Facilitated community outreach by conducting hands-on aeromodeling workshops to demonstrate UAV flight mechanics and foster an interest in STEM among high school students.

National Institute of Technology Karnataka

Project Intern

February 2025 - March 2025 (2 months)

Mangaluru

Collaborated on the development of a 1.5kg Unmanned Aerial Vehicle (UAV), focusing on a catapult launch system and a parachute-based safe recovery mechanism for post-mission retrieval.

Spearheaded the autonomy integration, configuring the UAV for automated mission execution and establishing reliable, real-time telemetry and communication with the Ground Control Station (GCS).

Engineered the Center of Gravity (CG) balancing, calculating and maintaining aerodynamic stability for both the standard flight configuration and the shifted weight dynamics post-parachute deployment.

Utilized MATLAB and Simulink to model flight dynamics, test control algorithms, and simulate the launch and landing mechanisms.

Education

Sahyadri College of Engineering & Management

Bachelor of Engineering - BE, Robotics and Automation Engineering · (August 2023 - June 2027)